

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1491

Melrose Electric 138 kV
October 15, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1491. The Interconnection Customer proposes a 363.4 MW solar photovoltaic facility interconnecting at the Melrose Electric 138 kV substation in the Kentucky area. The request was received on June 07, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 491 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$54,479,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Wyco – Melrose Electric 138.0 kV	Melros Electric – Mistersky 138.0 kV L	171.6	200	100.0
Mistersky – Melrose Electric 138.0 kV	Wyco – Melrose Electric 138.0 kV	147.0	233	100.0
Southaven – Mistersky 138.0 kV line	Wyco – Melrose Electric 138.0 kV	121.3	227	47.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wyco – Melrose Electric 138.0 kV	Rebuild line with higher-capacity conductor	\$12,195,000	100% (\$12,195,000)
Mistersky – Melrose Electric 138.0 kV	Rebuild line with higher-capacity conductor	\$36,974,000	100% (\$36,974,000)
Southaven – Mistersky 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,988,000	76% (\$5,310,000)
	Total allocated to project		\$54,479,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.